



Computer
Science

CSC380: Principles of Data Science

Probability 2

Xinchen Yu

Summary: calculating probabilities

2

- If we know that all outcomes are **equally likely**, we can use

We will use combinatorics
to do counting

$$P(E) = \frac{|E|}{|S|}$$

Number of elements
in event set

Number of possible
outcomes (e.g. 36)

- If $|E|$ is hard to calculate directly, we can try
 - the rules of probability
 - the Law of Total Probability, using an appropriate partition of sample space S

Rules of probability

- To recap and summarize:

Rules of Probability

- 1. Non-negativity:** All probabilities are between 0 and 1 (inclusive)
- 2. Unity of the sample space:** $P(S) = 1$
- 3. Complement Rule:** $P(E^C) = 1 - P(E)$
- 4. Probability of Unions:**
 - (a) In general, $P(E \cup F) = P(E) + P(F) - P(E \cap F)$*
 - (b) If E and F are disjoint, then $P(E \cup F) = P(E) + P(F)$*

- Conditional probability
- Probabilistic reasoning
 - contingency table
 - probability trees
- Bayes rule
- Independence of events
- Probability and combinatorics

Conditional Probability

Example: Seat Belts

		Child		
		Buck.	Unbuck.	Marginal
Parent	Buck.	0.48	0.12	0.60
	Unbuck.	0.10	0.30	0.40
Marginal		0.58	0.42	1.00

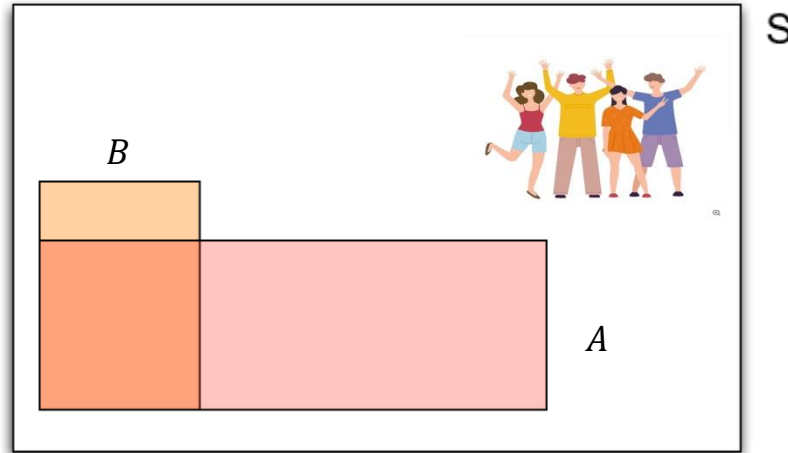
Table: Probability Estimates for Seat Belt Status

Suppose we pick a family from US at random:

- What is the probability of the event “Child is Buckled”?
- What should our new estimate be if we know that “Parent is Buckled”?

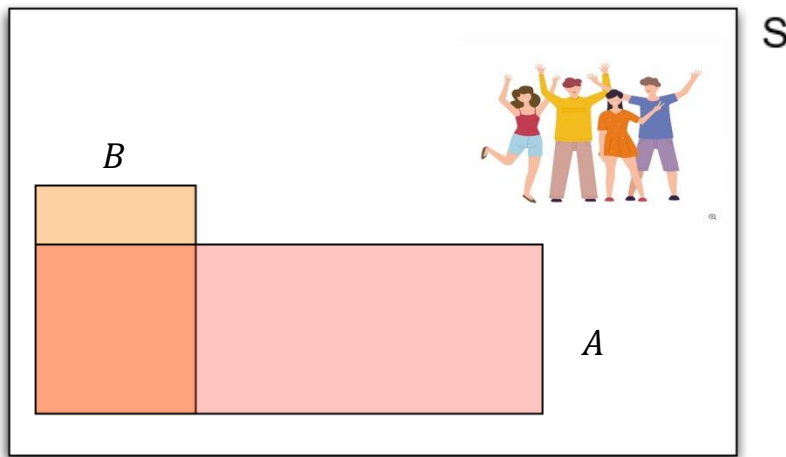
Absolute area

- A : antigen A present B : antigen B present
- What is the chance of B happening?



Relative area

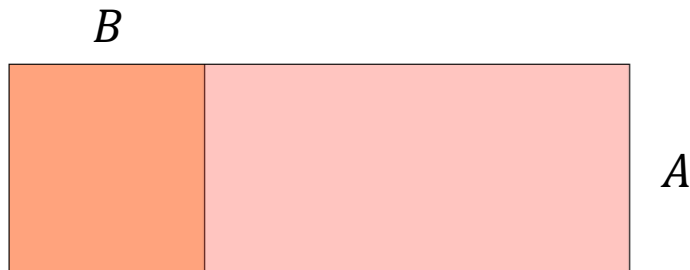
- A : antigen A present B : antigen B present
- Given that A happens, what is the chance of B happening?



- Another way to think about this:
 - Restricted to people with antigen A present, what is the fraction of those people with antigen B?

Relative area

- Let's zoom into people with antigen A present.



- It's just as if the sample space had shrunk to include only A
- Now, probabilities correspond to proportions of A
- What does the orange square represent in the original sample space?
 - $A \cap B$
- How would we find the probability of B given A ?

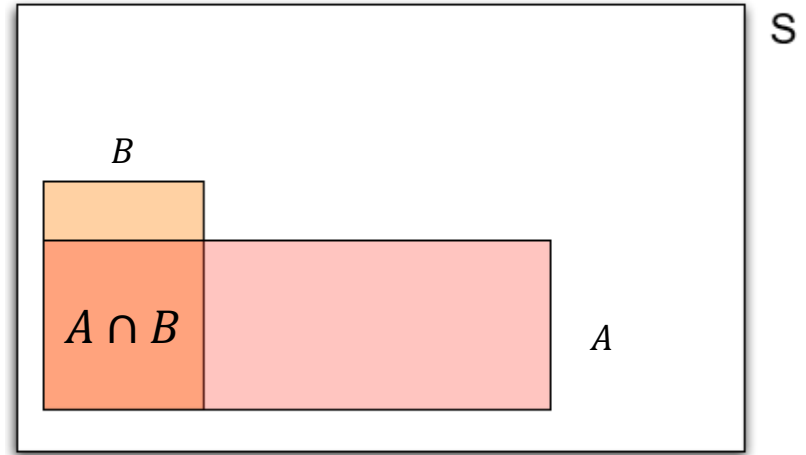
Conditioning changes the sample space

- Before we knew anything, anything in sample space S could occur.
- After we know A happened, we are only choosing from within A .
e.g., A : even numbers, B : get a 2
- The set A becomes our new sample space
- Instead of asking “In what proportion of S is B true?”, we now ask “In what proportion of A is B true?” e.g., $1/6$ vs $1/3$

Conditional Probability

- To find the conditional probability of B given A , consider the ways B can occur in the context of A (i.e., $A \cap B$), out of all the ways A can occur:

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$



Every Probability is a Conditional Probability

- We can consider the original probabilities to be conditioned on the event S : at first what we know is that “something in S ” occurs. E.g.

$$P(B) = P(B|S)$$

$$P(B \cap C) = P(B \cap C|S)$$

- $P(B|S)$ in words: what proportion of S does B happen?
- If we then learn that A occurs, A becomes our restricted sample space.
 $P(B|A)$ in words: what proportion of A does B happen?

When we have two events A and B...

- Conditional probability: $P(A|B)$, $P(A^c|B)$, $P(B|A)$ etc.
- Joint probability: $P(A, B)$ or $P(A^c, B)$ or ...
- Marginal probability: $P(A)$ or $P(A^c)$

Joint Probability and Conditional Probability

- We can rearrange $P(B | A) = \frac{P(A \cap B)}{P(A)}$ and derive:

The “Chain Rule” of Probability

For any events, A and B , the joint probability $P(A \cap B)$ can be computed as

$$P(A \cap B) = P(B|A) \times P(A)$$

Or, since $P(A \cap B) = P(B \cap A)$

$$P(A \cap B) = P(A|B) \times P(B)$$

Example: blood types

		Antigen B		Marginal
		Absent	Present	
Antigen A	Absent	0.44	0.10	0.54
	Present	0.42	0.04	0.46
Marginal		0.86	0.14	1.00

Table: Probability Estimates for U.S. Blood Types

- A : “presence of antigen A ”, and B : “presence of antigen B ”
- Suppose someone of an unknown blood type gets a test that reveals the presence of antigen A .
 - What is the chance that event A happens to them now?
 - What is the chance that event B happens to them now?

Example revisited: blood types

		Antigen B		Marginal
		Absent	Present	
Antigen A	Absent	0.44	0.10	0.54
	Present	0.42	0.04	0.46
Marginal		0.86	0.14	1.00

Table: Probability Estimates for U.S. Blood Types

- Suppose someone of an unknown blood type gets a test that reveals the presence of antigen A.

- What is $P(A | A)$?

$$P(A | A) = \frac{P(A \cap A)}{P(A)} = 1$$

- What is $P(B | A)$?

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{0.04}{0.46} = 0.087$$

Example revisited: Seat Belts

A: parent is buckled

C: child is buckled

		Child		Marginal
		Buck.	Unbuck.	
Parent	Buck.	0.48	0.12	0.60
	Unbuck.	0.10	0.30	0.40
Marginal		0.58	0.42	1.00

Table: Probability Estimates for Seat Belt Status

Suppose we pick a family from US at random:

- What is the probability of the event “Child is Buckled”? $P(C)$
- What should our new estimate be if we know that (“given that”) Parent is Buckled? $P(C | A)$

Example revisited: Seat Belts

A: parent is buckled

C: child is buckled

		Child		Marginal
		Buck.	Unbuck.	
Parent	Buck.	0.48	0.12	0.60
	Unbuck.	0.10	0.30	0.40
Marginal		0.58	0.42	1.00

Table: Probability Estimates for Seat Belt Status

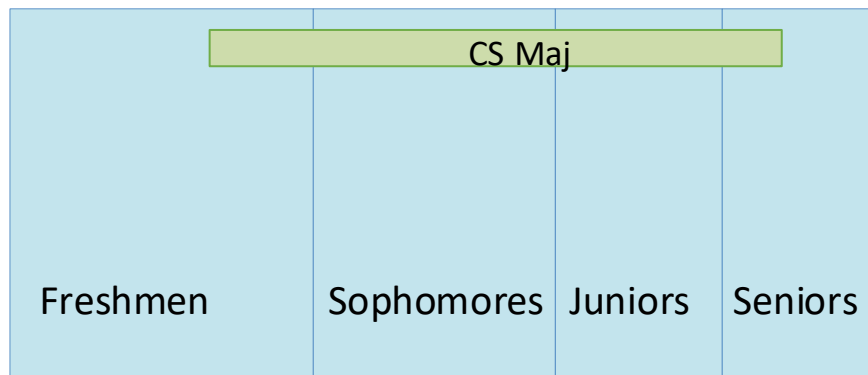
Suppose we pick a family from the US at random:

- $P(C) = 0.58$
- $P(C | A) = \frac{P(C \cap A)}{P(A)} = \frac{0.48}{0.60} = 0.8$ Larger than $P(C)$
- Suppose we see a buckled parent, it is much more likely that we see their child buckled

Law of Total Probability, revisited

Law of Total Probability Suppose $B_1, \dots, B_n$ form a partition of the sample space S . Then,

$$P(A) = P(A, B_1) + \dots + P(A, B_n)$$



Law of Total Probability, revisited

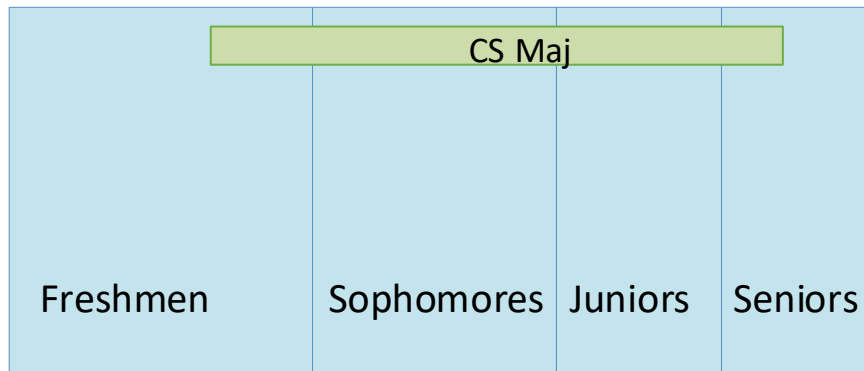
Expanding each $P(A, B_i) = \sum_n P(A | B_i)P(B_i)$, we have:

$$P(A) = \sum_{i=1} P(A | B_i)P(B_i)$$

A : student in CS major

B_i : student in class year i

$P(A | B_i)$ The fraction of CS major in class year i



Law of Total Probability, revisited

Example Suppose UA has an equal number of students in the 4 class years, and the fraction of CS major in these 4 class years are 10%, 10%, 20%, 80% respectively. What is fraction of CS majors?

- $P(B_1) = P(B_2) = P(B_3) = P(B_4) = 0.25$
- $P(C | B_1) = 0.1, \dots, P(C | B_4) = 0.8$
- Calculate $P(C)$ by:

$$P(C) = \sum_{i=1}^4 P(C | B_i)P(B_i) = 30\%$$

Probabilistic reasoning

Probabilistic reasoning

- We have some prior belief of an event A happening
 - $P(A)$, prior probability
 - e.g. me infected by COVID
- We see some new evidence B
 - e.g. I test COVID positive
- How does seeing B affect our belief about A ?
 - $P(A | B)$, posterior probability



Another example: detector

A store owner discovers that some of her employees have taken cash. She decides to use a detector to discover who they are.

- Suppose that 10% of employees stole.
- The detector buzzes 80% of the time that someone stole, and 20% of the time that someone not stole
- Is the detector reliable? In other words, if the detector buzzes, what's the probability that the person did stole?

H: employee not stole

B: lie detector buzzes

Another example: detector

- Suppose that 10% of employees stole.

H: employee did not stole $P(H) = 0.9$

- The detector buzzes 80% of the time that someone stoles, and 20% of the time that someone not stole.

$$P(B \mid H^c) = 0.8$$

B: lie detector buzzes

$$P(B \mid H) = 0.2$$

- If the detector buzzes, what's the probability that the person stole?

$$P(H^c \mid B)$$

Detector analysis: Probability table

		Detector result		
		Pass	Buzz (B)	
Employee	Not stole (H)			
	Stole			
	Marginal			

$$P(H) = 0.9$$

$$P(B \mid H^c) = 0.8$$

$$P(B \mid H) = 0.2$$

Detector analysis: Probability table

$$P(H, B) = P(H) \cdot P(B | H) = 0.9 \times 0.2 = 0.18$$

		Detector result		
		Pass (B^C)	Buzz (B)	Marginal
Employee	Not stole (H)		0.18	0.9
	Stole (H^C)			0.1
	Marginal			

$$P(H) = 0.9$$

$$P(B | H^C) = 0.8$$

$$P(B | H) = 0.2$$

Detector analysis: Probability table

$$P(H) = P(H, B) + P(H, B^c) = 0.9$$

		Detector result		
		Pass (B^c)	Buzz (B)	Marginal
Employee	Not stole (H)	0.72	0.18	0.9
	Stole (H^c)			0.1
	Marginal			

$$P(H) = 0.9$$

$$P(B \mid H^c) = 0.8$$

$$P(B \mid H) = 0.2$$

Detector analysis: Probability table

		Detector result		Marginal
		Pass (B^C)	Buzz (B)	
Employee	Not stole (H)	0.72	0.18	0.9
	Stole (H^C)	0.02	0.08	0.1
	Marginal	0.74	0.26	1

$$P(H) = 0.9$$

$$P(B \mid H^C) = 0.8$$

$$P(B \mid H) = 0.2$$

Detector analysis: Probability table

		Detector result		
		Pass (B^C)	Buzz (B)	Marginal
Employee	Not stole (H)	0.72	0.18	0.9
	Stole (H^C)	0.02	0.08	0.1
Marginal		0.74	0.26	1

- We have the full probability table. Can we calculate $P(H^C | B)$? Yes!

$$P(H^C | B) = \frac{P(H^C, B)}{P(B)} = \frac{0.08}{0.26} = 0.307$$

It seems like the detector is not very reliable...